Electronic Journal of Statistics
ISSN: 1935-7524

When Your Model Stops Working: Anytime-Valid Calibration Monitoring

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Abstract: Practitioners monitoring deployed probabilistic models face a fundamental trap: any fixed-sample test applied repeatedly over an unbounded stream will eventually raise a false alarm, even when the model remains perfectly stable. Existing methods typically lack formal error guarantees, conflate alarm time with changepoint location, and monitor indirect signals that do not fully characterise calibration. We present PITMonitor, an *anytime-valid* calibration-specific monitor that detects distributional shifts in probability integral transforms via a mixture e-process, providing Type I error control over an unbounded monitoring horizon and principled changepoint estimation. On `river`'s FriedmanDrift benchmark, PITMonitor achieves detection rates competitive with the strongest baselines across all three scenarios, although detection delay is substantially longer under local drift. Code is available at `https://github.com/tristan-farran/pitmon`.

MSC2020 subject classifications: Primary 62L10; secondary 62G10.
Keywords and phrases: anytime-valid inference, calibration monitoring, e-values, e-processes, regime shifts, sequential hypothesis testing, changepoint detection, concept drift.

1. Introduction

Probabilistic models deployed in production face a fundamental challenge: the world changes. Across domains, from medicine to finance, models encounter regime shifts and concept drift which can cause calibration to degrade drastically with consequential effects downstream. Modern neural networks, for instance, can improve classification accuracy while simultaneously becoming less calibrated (Guo et al., 2017). Practitioners must therefore monitor their models continuously, seeking to answer two questions: *has* the world changed, and if so, *when*? In practice, this monitoring often relies on ad-hoc procedures such as periodic recalibration schedules, rolling-window hypothesis tests, or threshold-based alerts on summary metrics.

These approaches suffer from a fundamental statistical problem: *they do not control the false alarm rate over continuous monitoring*. A practitioner who checks calibration daily with a $p < 0.05$ threshold will, over a year of monitoring, almost certainly observe spurious alarms even if calibration remains stable. Classical hypothesis tests assume a fixed sample size determined before seeing data; continuous monitoring violates this assumption.

More principled alternatives are provided by online drift detectors, such as those implemented in the `river` library (Montiel et al., 2021). Classical detectors such as DDM and HDDM are lightweight and often effective at detecting abrupt changes, but rely on heuristic thresholds or fixed-sample statistical arguments that do not provide false alarm guarantees under continuous monitoring. ADWIN (Bifet and Gavaldà, 2007) improves on fixed-window methods by adapting its window size to bound the false alarm probability *within a single window*, but this per-window guarantee does not extend to the stream level: as the stream grows, the number of implicit tests accumulates without bound, inflating the false alarm rate.

More fundamentally, these methods typically monitor generic signals such as error rates or mean changes in residuals rather than calibration-specific quantities, with finer notions of miscalibration such as overconfidence with unchanged accuracy potentially going undetected.

We propose PITMonitor, an anytime-valid calibration monitor with five key properties:

1. **Anytime-valid false alarm control:** we prove that $\mathbb{P}(\text{ever alarm} \mid H_0) \leq \alpha$ for all time, without requiring a pre-specified monitoring horizon or stopping rule.
2. **Calibration-specificity:** we target the full calibration relationship between model and data, rather than indirect signals that may miss important aspects of miscalibration.
3. **No baseline period required:** unlike methods requiring a "clean" reference distribution or initialisation window, PITMonitor works from the first observation.
4. **Practical efficiency:** the method runs in $O(1)$ per update of the mixture e-process, plus $O(\log n)$ for rank maintenance via a `SortedList` (Jenks, 2014), with $O(n)$ memory.
5. **Distribution-free:** no assumptions are made about the pre- or post-change distributions.

2. Related Work

Calibration Assessment

Classical calibration metrics include expected calibration error (Guo et al., 2017), reliability diagrams (DeGroot and Fienberg, 1983), and proper scoring rules (Gneiting and Raftery, 2007). PITs have been used for static forecast evaluation in econometrics (Diebold, Gunther and Tay, 1998) and weather forecasting (Gneiting and Katzfuss, 2014). These both provide static calibration assessment but do not address sequential monitoring with false alarm control.

Distribution Shift Detection

Methods for detecting distribution shift include two-sample tests (Rabanser, Günnemann and Lipton, 2019) and domain classifiers (Lipton, Wang and Smola, 2018). These methods primarily address changes in the input or label distributions, while our work focuses on the *output* side: identifying when predicted probabilities no longer match outcome frequencies.

Sequential Calibration Testing

Arnold, Henzi and Ziegel (2023) proposed e-values for testing forecast calibration, focusing on whether PITs are Uniform$(0, 1)$. PITMonitor differs in two ways. First, we test for *changes* in the PIT process rather than absolute uniformity and thus alarm only on shifts in calibration — the more natural null when some level of miscalibration is inevitable. Second, we locate *when* calibration changed, rather than only establishing *whether* the model is miscalibrated.

E-values

E-values support anytime-valid hypothesis testing under optional stopping (Vovk and Wang, 2021; Ramdas et al., 2023; Grünwald, de Heide and Koolen, 2024; Shin, Ramdas and Rinaldo, 2024). PITMonitor builds on this framework, constructing a mixture e-process to detect shifts in the PIT distribution and inheriting anytime-valid false-alarm guarantees (Theorem 1).

3. Background

3.1. Probability Integral Transforms

A probabilistic model with predictive CDF $\hat{F}$ for $Y \sim P$ is *perfectly calibrated* if $\hat{F} = P$. *Miscalibration*, meaning divergence of predicted probabilities from outcome frequencies, can manifest as systematic bias, over- or under-confidence, misassigned tail risks, and more. The *probability integral transform* (PIT) $U = \hat{F}(Y)$ unifies all of these into a single object computable from predictions and observations alone.

Since U has distribution $P \circ \hat{F}^{-1}$, it encodes the full calibration relationship between model and data-generating process: every discrepancy between $\hat{F}$ and P leaves a signature in the distribution of U, and conversely every feature of the PIT distribution corresponds to a specific calibration property. Perfect calibration corresponds to $U \sim \text{Uniform}(0, 1)$ (Dawid, 1984), but the shape of the PIT distribution is informative more broadly: a U-shaped density indicates underdispersion, a hump near the center overdispersion, and a triangular density systematic location bias (Gneiting, Balabdaoui and Raftery, 2007). Calibration changes if and only if the PIT distribution changes, making the latter a natural monitoring target. We therefore test:

- $\mathbf{H_0}$: $U_t \overset{\text{i.i.d.}}{\sim} F$ for all t
- $\mathbf{H_1}$: $\exists \tau \geq 1$ such that $(U_t)_{t \leq \tau} \overset{\text{i.i.d.}}{\sim} F$ and $(U_t)_{t > \tau}$ is not i.i.d. from F.

As a result, PITMonitor is able to target any change in the calibration relationship, without alarming on static miscalibration. [1] The null is broad, encompassing both perfectly calibrated models ($F = \text{Uniform}(0, 1)$) and miscalibrated yet stable ones ($F \neq \text{Uniform}(0, 1)$), while the alternative captures any change in calibration, whether abrupt or gradual, local or global.

3.2. Conformal P-values

To detect distribution shifts sequentially, we employ *conformal p-values from ranks* (Vovk, Gammerman and Shafer, 2005). Given observations $U_1, \ldots, U_t$, define the rank of U_t as:

$$R_t = \#\{s \leq t : U_s \leq U_t\} \tag{1}$$

Proposition 1 (Rank Uniformity Under H_0)**.** *If $U_1, \ldots, U_t \overset{i.i.d.}{\sim} F$ with F continuous,*[2] *then* $R_t \sim \text{Unif}\{1, \ldots, t\}$.

Proof. Since F is continuous, ties occur with probability zero and all $t!$ orderings of $(U_1, \ldots, U_t)$ are equally likely. Exactly $(t-1)!$ place U_t at rank r, giving $\mathbb{P}(R_t = r) = 1/t$. □

Since R_t is discrete, we add an independent jitter V_t to obtain an exactly uniform p-value:

$$p_t = \frac{R_t - 1 + V_t}{t}, \quad V_t \sim \text{Uniform}(0, 1) \tag{2}$$

Under H_1, new observations no longer fall uniformly across the reference sample: they are systematically displaced toward regions where the new law concentrates more or less mass than F. Intuitively, this displacement biases the ranks R_t and hence the p-values p_t away from uniformity, making them informative signals of change. A formal characterisation of the post-change p-value behaviour is non-trivial, as it depends on the entire pre-change history and the nature of the shift, so it is left for future work (Section 7).

[1] The i.i.d. idealisation is rare in practice but provides a tractable foundation; see 6.

[2] If F is discretized or rounded, randomized tie-breaking can be used to preserve uniformity.

3.3. E-values

An *e-value* is a non-negative random variable satisfying $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \leq 1$ under the null hypothesis. A key property of e-values is that they can be composed multiplicatively while maintaining validity under the null (Ramdas et al., 2023). The resulting product is an *e-process*, accumulating evidence from time τ onwards:

$$M_t^{(\tau)} = \prod_{s=\tau}^{t} e_s \tag{3}$$

Under the alternative, e-values are constructed so that $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] > 1$. Therefore,

$$\mathbb{E}[M_t^{(\tau)} \mid \mathcal{F}_{t-1}] = M_{t-1}^{(\tau)} \cdot \mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \begin{cases} \leq M_{t-1}^{(\tau)} & \text{under } H_0, \\ > M_{t-1}^{(\tau)} & \text{under } H_1. \end{cases} \tag{4}$$

Hence, $M_t^{(\tau)}$ is a non-negative supermartingale under H_0, yet grows without bound in expectation under H_1. This motivates detecting a shift by monitoring whether the process exceeds a threshold. By Ville's inequality (Ville, 1939), $\mathbb{P}(\sup_{t \geq \tau} M_t^{(\tau)} \geq 1/\alpha) \leq \alpha$ under H_0, so the threshold $1/\alpha$ controls the false alarm probability at level α over an unbounded horizon (as we formalise in Theorem 1). We therefore raise an alarm at the first time t such that $M_t^{(\tau)} \geq 1/\alpha$.

4. Method

4.1. E-values via Density Betting

To construct e-values from p-values, we use the betting framework of Ramdas et al. (2023). At time t, before observing p_t, we choose a density function $\hat{f}_t$ satisfying $\int_0^1 \hat{f}_t(p)\, dp = 1$. Interpreting this as a wager against H_0, we pay one unit up front and receive payoff $\hat{f}_t(p_t)$ after p_t is revealed, giving the one-step e-value $e_t = \hat{f}_t(p_t)$.

Proposition 2 (Density Betting Yields Valid E-values)**.** *Let $\hat{f} : [0,1] \to [0,\infty)$ be a density function satisfying $\int_0^1 \hat{f}(p)\, dp = 1$. If $p \sim \text{Uniform}(0,1)$, then $e = \hat{f}(p)$ satisfies $\mathbb{E}[e] = 1$.*

Proof. Since $p \sim \text{Uniform}(0,1)$, $\mathbb{E}[e] = \mathbb{E}[\hat{f}(p)] = \int_0^1 \hat{f}(p)\, dp = 1$. □

To bet on p-values adaptively, we use a nonparametric plug-in histogram density estimator which places mass proportional to where past p-values concentrated (for the broader plug-in martingale lineage, see Fedorova et al. (2012)). Letting c_b count past p-values in bin b and B be the number of bins:

$$\hat{f}(p) = B \cdot \frac{c_b}{\sum_j c_j} \quad \text{for } p \in \text{bin } b \tag{5}$$

The histogram is initialized with pseudocounts $c_b = 1$ for all b and updated *after* computing e_t, ensuring $\hat{f}$ is a valid $\mathcal{F}_{t-1}$-measurable density from the first observation. Since p-values are Uniform$(0,1)$ under H_0 by Proposition 1, this produces a valid e-value by Proposition 2.

Under H_1, the distribution shifts and p-values are non-uniform as described in Section 3.2. As shifted p-values fall more frequently in certain bins, the histogram learns their concentration pattern and bets accordingly, yielding $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] > 1$ whenever deviations are sufficiently persistent, with larger shifts producing faster e-process growth (as formalised under idealised conditions in Proposition 3).

Proposition 3 (Finite-Sample Mean Gain)**.** *Let $\tau < t$ be a changepoint, and $n := t - \tau$ be the number of post-shift observations preceding time t. Define:*

- $A_b := \#\{s < \tau : p_s \in \text{bin } b\}$, $m := \sum_{b=1}^{B} A_b = \tau - 1$ *(pre-shift counts),*
- $C_b := \#\{\tau \le s \le t-1 : p_s \in \text{bin } b\}$, $\sum_{b=1}^{B} C_b = n$ *(post-shift counts),*
- $\hat{q}_b := (1 + A_b)/(B + m)$ *(smoothed pre-shift density).*

Assume post-shift p-values are i.i.d. with bin probabilities $\theta = (\theta_1, \dots, \theta_B)$. Define

$$\gamma := B \sum_{b=1}^{B} \theta_b \hat{q}_b - 1, \qquad \delta := \sum_{b=1}^{B} \theta_b^2 - \frac{1}{B}, \tag{6}$$

where γ measures the alignment of θ with the pre-shift prior $\hat{q}$ (the factor B normalizes so that $\gamma = 0$ when $\hat{q}_b = 1/B$, i.e. when the pre-shift histogram is uninformative), and $\delta \ge 0$ measures shift intensity, vanishing iff θ is uniform. Then, conditioning on $A = (A_1, \dots, A_B)$:

$$\mathbb{E}[e_t \mid A] = 1 + \frac{B+m}{B+m+n}\gamma + \frac{n}{B+m+n} B\delta. \tag{7}$$

Proof. The estimator and e-value are

$$\hat{\theta}_{t,b} = \frac{1 + A_b + C_b}{B+m+n}, \qquad e_t = B\hat{\theta}_{t,b_t}.$$

Since p_t independently falls in bin b with probability θ_b,

$$\mathbb{E}[e_t \mid A] = \frac{B}{B+m+n} \sum_{b=1}^{B} \theta_b \, \mathbb{E}[1 + A_b + C_b \mid A].$$

Under the i.i.d. post-shift model $\mathbb{E}[C_b \mid A] = n\theta_b$, so

$$\mathbb{E}[e_t \mid A] = \frac{B}{B+m+n} \Big[\underbrace{\sum_{b=1}^{B} \theta_b (1 + A_b)}_{(B+m)(\gamma+1)/B} + n \underbrace{\sum_{b=1}^{B} \theta_b^2}_{\delta + 1/B} \Big].$$

Expanding and collecting gives (7). □

Proposition 3 shows the mean gain interpolating between two regimes: for small n the prior-alignment term γ dominates, while as $n \to \infty$ the data weight approaches one and $\mathbb{E}[e_t \mid A] \to 1 + B\delta$ The transition speed is governed by n relative to $B + m$, with longer pre-shift histories delaying convergence.

Since γ can be negative (as low as $-m/(B+m)$ when the prior is adversarially misaligned with the shift), $\mathbb{E}[e_t \mid A]$ may initially fall below 1, meaning early e-values are expected to be less than one and dilute the e-process. From (7), the crossover $\mathbb{E}[e_t \mid A] > 1$ requires $(B+m)\gamma + Bn\delta > 0$, i.e.

$$n > n^* := \frac{(B+m)|\gamma|}{B\delta} \qquad (\gamma < 0). \tag{8}$$

This warmup length grows with the severity of misalignment $|\gamma|$ and shrinks with shift intensity δ: a larger shift overwhelms a bad prior faster. Since $|\gamma| \le m/(B+m)$, the worst-case warmup is bounded by $m/(B\delta)$, which itself shrinks as δ grows. When $\gamma \ge 0$ the prior is aligned with the shift and $\mathbb{E}[e_t \mid A] > 1$ immediately.

Remark 1 (Transience and detection window). Although $p_{\tau+k}$ is non-uniform for every finite k, the degree of non-uniformity decays as post-change observations accumulate in the reference. The contamination fraction of the reference at time $\tau + k$ is

$$\lambda_k = \frac{k-1}{\tau + k - 1},$$

rising from 0 at $k = 1$ to $1/2$ at $k \sim \tau$ and approaching 1 as $k \to \infty$. As $\lambda_k \to 1$, the effective reference CDF converges to G, so $G(U_{\tau+k}) \sim \text{Uniform}(0, 1)$ and non-uniformity vanishes. The detection window is therefore roughly τ samples wide: evidence must accumulate before the reference becomes too contaminated to sustain a signal.

This directly bounds the power analysis in Section 4.1. The shift intensity $\delta = \sum_b \theta_b^2 - 1/B$ in Proposition 3 is implicitly time-varying: the effective θ starts near $G(F^{-1}(\cdot))$ and decays toward uniform as λ_k grows, so δ shrinks and the mean e-value gain decreases over the post-change window.

Proposition **??** establishes that $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] > 1$ asymptotically after a shift, under the assumption that post-change p-values are i.i.d. non-uniform. In Algorithm 1, however, the conformal p-value p_t is computed from the rank of U_t against the *entire* past sequence. As quantified in Remark 1, this reference absorbs post-change observations at rate $\lambda_k = (k-1)/(\tau + k - 1)$, so the non-uniformity driving the mean gain is transient and spans roughly τ post-change samples. Proposition **??** should therefore be read as characterizing the density estimator's behaviour under an exogenously supplied non-uniform p-value stream rather than a permanent-regime guarantee for the self-normalizing conformal construction.

4.2. The Mixture E-process

A key challenge in constructing our e-process is that the true changepoint time T is unknown. An e-process starting at τ would be sensitive to drift beginning at τ but would miss earlier changes, while one starting too early would accumulate excess noise that dilutes its power. Since a weighted mixture of e-processes is itself a valid e-process by linearity of expectation (Ramdas et al., 2023), we address the unknown changepoint time by maintaining one over all possible start times:

$$M_t = \sum_{\tau=1}^{t} w_\tau \cdot M_t^{(\tau)} \tag{9}$$

where $M_t^{(\tau)} = \prod_{s=\tau}^{t} e_s$ is defined for $\tau \leq t$. Since each component $M_t^{(\tau)}$ forms a valid e-process sensitive to drift beginning at τ, the mixture is sensitive to changepoints at all times. We use $w_\tau = 1/(\tau(\tau+1))$ as this sequence satisfies $\sum_{\tau=1}^{\infty} w_\tau = 1$ exactly, enabling an efficient recursion that avoids maintaining separate products for each τ:

Proposition 4 (Efficient Recursion)**.** *The mixture e-process satisfies:*

$$M_t = e_t \cdot (M_{t-1} + w_t) \tag{10}$$

Proof. Expand the definition:

$$M_t = \sum_{\tau=1}^{t} w_\tau \cdot M_t^{(\tau)} \tag{11}$$

$$= \sum_{\tau=1}^{t-1} w_\tau \cdot e_t \cdot M_{t-1}^{(\tau)} + w_t \cdot e_t \tag{12}$$

$$= e_t \left(\sum_{\tau=1}^{t-1} w_\tau \cdot M_{t-1}^{(\tau)} + w_t \right) \tag{13}$$

$$= e_t (M_{t-1} + w_t) \tag{14}$$

□

This recursion enables a constant-time update of the mixture per observation, with additional cost from rank maintenance, while avoiding the cost of maintaining or updating all e-processes separately.

Remark 2 (Mixture vs. Component Guarantees). Proposition **??** establishes almost-sure detection for the component e-process $M_t^{(\tau)}$ started at a *known* changepoint τ. Since the mixture $M_t = \sum_\tau w_\tau M_t^{(\tau)}$ is a positive linear combination of components rather than a product of e-values, its power properties require separate treatment.

Proposition 5 (Eventual Alarm under H_1 – Mixture)**.** *Under the conditions of Proposition* **??**, *the mixture e-process satisfies, for any* $\alpha \in (0, 1)$,

$$\mathbb{P}\left(\exists\, t \geq T : M_t \geq \frac{1}{\alpha} \right) = 1.$$

Proof. Since all terms of $M_t = \sum_{\tau=1}^{t} w_\tau M_t^{(\tau)}$ are non-negative, $M_t \geq w_T M_t^{(T)}$ for all $t \geq T$. By Proposition **??**, $M_t^{(T)} \to \infty$ almost surely, so $M_t \to \infty$ almost surely. Hence for any finite threshold $1/\alpha$, there exists (a.s.) some t with $M_t \geq 1/\alpha$. □

The weight $w_T = 1/(T(T+1))$ also governs detection sensitivity. Since $M_t \geq w_T M_t^{(T)}$, the mixture fires whenever the component $M_t^{(T)}$ reaches the scaled threshold $1/(\alpha w_T) = T(T+1)/\alpha$. The cost of changepoint uncertainty therefore appears as a multiplicative factor $T(T+1)$ on the effective threshold: later changepoints require proportionally stronger evidence, but the penalty grows only quadratically in T rather than exponentially.

4.3. Type I Error Control

Theorem 1 (Anytime-Valid False Alarm Control)**.** *Under* H_0, *PITMonitor satisfies:*

$$\mathbb{P}\left(\sup_{t \geq 1} M_t \geq \frac{1}{\alpha} \right) \leq \alpha \tag{15}$$

Proof. The mixture $M_t = \sum_{\tau=1}^{t} w_\tau M_t^{(\tau)}$ is defined with $M_t^{(\tau)}$ only for $\tau \leq t$. To apply Ville's inequality, we work with an extended process defined for all $t \geq 0$. For each $\tau \geq 1$ define:

$$\widetilde{M}_t^{(\tau)} = \begin{cases} 1, & t < \tau, \\ \prod_{s=\tau}^{t} e_s, & t \geq \tau. \end{cases} \tag{16}$$

Define the full mixture over all $\tau \geq 1$:

$$\widetilde{M}_t = \sum_{\tau=1}^{\infty} w_\tau \, \widetilde{M}_t^{(\tau)}. \tag{17}$$

Since $\widetilde{M}_t^{(\tau)} = 1$ for $\tau > t$, we can expand:

$$\widetilde{M}_t = \sum_{\tau=1}^{t} w_\tau \prod_{s=\tau}^{t} e_s + \sum_{\tau=t+1}^{\infty} w_\tau \tag{18}$$

$$= M_t + \sum_{\tau=t+1}^{\infty} \frac{1}{\tau(\tau+1)}. \tag{19}$$

With $w_\tau = \frac{1}{\tau(\tau+1)}$, the tail telescopes:

$$\sum_{\tau=t+1}^{\infty} \frac{1}{\tau(\tau+1)} = \sum_{\tau=t+1}^{\infty} \left(\frac{1}{\tau} - \frac{1}{\tau+1} \right) = \frac{1}{t+1}.$$

Hence

$$M_t \leq M_t + \frac{1}{t+1} = \widetilde{M}_t, \tag{20}$$

$$\therefore \left\{ \sup_{t \geq 1} M_t \geq \frac{1}{\alpha} \right\} \subseteq \left\{ \sup_{t \geq 1} \widetilde{M}_t \geq \frac{1}{\alpha} \right\}. \tag{21}$$

Since each $(\widetilde{M}_t^{(\tau)})_{t \geq 0}$ is a non-negative supermartingale with $\widetilde{M}_0^{(\tau)} = 1$. By Tonelli's theorem,

$$\mathbb{E}[\widetilde{M}_t \mid \mathcal{F}_{t-1}] = \sum_{\tau=1}^{\infty} w_\tau \, \mathbb{E}[\widetilde{M}_t^{(\tau)} \mid \mathcal{F}_{t-1}] \tag{22}$$

$$\leq \sum_{\tau=1}^{\infty} w_\tau \, \widetilde{M}_{t-1}^{(\tau)} = \widetilde{M}_{t-1}, \tag{23}$$

so $(\widetilde{M}_t)$ is a non-negative supermartingale with $\widetilde{M}_0 = \sum_{\tau=1}^{\infty} w_\tau = 1$. Finally, applying Ville's inequality,

$$\mathbb{P}\left(\sup_{t \geq 1} M_t \geq \frac{1}{\alpha} \right) \leq \mathbb{P}\left(\sup_{t \geq 1} \widetilde{M}_t \geq \frac{1}{\alpha} \right) \leq \alpha. \tag{24}$$

□

4.4. Changepoint Estimation

After an alarm at time T, we estimate the changepoint location by selecting the split that best explains the post-split p-values as non-uniform. For each candidate $s \in \{1, \ldots, T-1\}$, we compute a split-specific histogram $\mathbf{n}^{(s)} = (n_1^{(s)}, \ldots, n_B^{(s)})$, where $n_b^{(s)}$ counts p-values in bin b and $N^{(s)} = \sum_b n_b^{(s)} = T - s$, and evaluate the segment $(p_{s+1}, \ldots, p_T)$ under two hypotheses:

- $\mathbf{H_0^{(s)}}$: p-values are Uniform$(0, 1)$, so each of B bins receives probability $1/B$.

- $\mathbf{H_1^{(s)}}$: bin probabilities are unknown, with a symmetric Dirichlet prior $\mathrm{Dir}(\kappa, \ldots, \kappa)$.[3]

We use a Bayes factor rather than a likelihood ratio because H_1 would otherwise trivially outperform H_0 by overfitting, since it contains H_0 as a special case. Averaging over the Dirichlet prior penalizes H_1 for the probability mass it wastes on configurations the data does not support. We set $\kappa = 1/2$, since it is invariant to the reparametrization of bin boundaries (Jeffreys, 1961).

Under H_0 every bin has probability $1/B$, so the multinomial log-likelihood reduces to:

$$\log p(\mathbf{n}^{(s)} \mid H_0^{(s)}) = \log \frac{N^{(s)}!}{\prod_b n_b^{(s)}!} - N^{(s)} \log B \tag{25}$$

Under H_1, the probabilities are unknown so we integrate them out over the Dirichlet prior:

$$\begin{aligned} \log p(\mathbf{n}^{(s)} \mid H_1^{(s)}) = \log \frac{N^{(s)}!}{\prod_b n_b^{(s)}!} &+ \log \Gamma(B\kappa) - \log \Gamma(N^{(s)} + B\kappa) \\ &+ \sum_{b=1}^{B} \left[\log \Gamma(n_b^{(s)} + \kappa) - \log \Gamma(\kappa)\right] \end{aligned} \tag{26}$$

Since the combinatorial factor appears in both likelihoods, it cancels in the log Bayes factor:

$$\begin{aligned} \log \mathrm{BF}(s) &= \log p(\mathbf{n}^{(s)} \mid H_1^{(s)}) - \log p(\mathbf{n}^{(s)} \mid H_0^{(s)}) \\ &= \log \Gamma(B\kappa) - \log \Gamma(N^{(s)} + B\kappa) \\ &\quad + \sum_{b=1}^{B} \left[\log \Gamma(n_b^{(s)} + \kappa) - \log \Gamma(\kappa)\right] \\ &\quad + N^{(s)} \log B \end{aligned} \tag{27}$$

We then identify the changepoint simply as $\hat{\tau} = \arg\max_s \log \mathrm{BF}(s)$.[4]

This localization capability is absent from all `river` baselines, which expose only a binary alarm flag. PITMonitor thus enables practitioners not just to detect that drift has occurred, but to identify approximately how far back model outputs may have been affected by the shift - directly actionable for deciding how much historical inference to distrust or recompute.

4.5. Complete Algorithm

5. Experiments

We evaluate PITMonitor on the FriedmanDrift benchmark, a standard regression task for evaluating concept drift detectors under controlled conditions, comparing against the seven stream-drift detectors included in `river`. Although FriedmanDrift induces covariate drift, feeding shifted inputs to a held-fixed model changes the PIT distribution, so covariate drift directly becomes calibration drift from the monitor's perspective.

[3] It should be noted that this is a heuristic approximation since the p_t sequence carries dependence across time.

[4] A natural alternative is the start time of the largest e-process, $\arg\max_\tau M_T^{(\tau)}$, but following from Proposition 3, we expect $e_s \approx 1$ for some time after a changepoint, biasing this alternative estimator toward later start times.

Algorithm 1 PITMonitor

Require: Significance level α, number of bins B
Initialize: $M_0 \leftarrow 0$, histogram counts $c_1, \ldots, c_B \leftarrow 1$
for $t = 1, 2, \ldots$ **do**
  Observe PIT $U_t \in [0, 1]$
  Insert U_t into sorted list; compute rank R_t
  Sample $V_t \sim \text{Uniform}(0, 1)$
  $p_t \leftarrow (R_t - 1 + V_t)/t$ ▹ Conformal p-value
  $b \leftarrow \min(\lfloor p_t \cdot B \rfloor + 1, B)$ ▹ Integer bin count
  $e_t \leftarrow B \cdot c_b / \sum_{j=1}^{B} c_j$ ▹ E-value from density
  $c_b \leftarrow c_b + 1$ ▹ Histogram update
  $w_t \leftarrow 1/(t \cdot (t + 1))$ ▹ Mixture weight
  $M_t \leftarrow e_t \cdot (M_{t-1} + w_t)$ ▹ Mixture e-process
  if $M_t \geq 1/\alpha$ **then** ▹ Threshold
    return ALARM at time t
  end if
end for

5.1. Setup

Dataset and drift scenarios. FriedmanDrift (Montiel et al., 2021) is a synthetic regression stream with 10 input features (x_0–x_9). We evaluate three drift types that represent qualitatively different distribution changes:

- **GRA** (Global Recurring Abrupt): All features change simultaneously at an abrupt onset, representing a rapid regime shift.
- **GSG** (Global Slow Gradual): The change spreads linearly across all features over a 500-sample transition window, representing gradual covariate shift.
- **LEA** (Local Expanding Abrupt): Drift starts on a subset of features and expands to include more over time, representing localized distribution change.

Stream layout. A training stream consisting of $n_{\text{train}} = 10{,}000$ pre-drift samples is generated, along with 10,000 independent test streams containing $n_{\text{stable}} = 2{,}500$ pre-drift monitoring samples that define the null-hypothesis window for FPR estimation, and $n_{\text{post}} = 2{,}500$ post-drift samples for TPR estimation. The drift onset occurs at the boundary between the pre- and post-drift segments.

Predictive model. We train a feedforward neural network outputting a Gaussian predictive distribution $\mathcal{N}(\mu_t, \sigma_t^2)$ for each input. The network has 3 hidden layers of 128 units with SiLU activations. Inputs and targets are standardized using per-feature means and standard deviations fitted on the training set. Training uses mini-batches of size 256, the Adam optimizer with initial learning rate 3×10^{-4}, a cosine annealing learning rate schedule, and 500 epochs. The model achieves $R^2 = 0.96$ on a held-out pre-drift test set, with an expected calibration error (ECE) of 0.01, confirming it is well-specified and calibrated before monitoring begins. The model is trained once and subsequently held fixed across all 10,000 Monte Carlo trials.

PIT construction. For each monitoring sample (x_t, y_t) the PIT is:

$$U_t = \Phi\left(\frac{y_t - \mu_t}{\sigma_t}\right) \tag{28}$$

where μ_t, σ_t are the predicted mean and standard deviation and Φ is the standard normal CDF.

Detector settings. All `river` baselines are run with their library-default parameters. This reflects an out-of-the-box comparison, avoids detector-specific tuning in favour of any particular method, and sidesteps the need for held-out null data or advance knowledge of the monitoring window length to align parameters with a target stream-level FPR.

- **PITMonitor**: Significance level $\alpha = 0.05$, number of bins $B = 100$.
- **Continuous-input baselines (ADWIN, KSWIN, PageHinkley)**: library-default parameters.
- **Binary-input baselines (DDM, EDDM, HDDM_A, HDDM_W)**: library-default parameters. These detectors require a binary error signal; we binarize via $b_t = \mathbf{1}[|r_t| > \theta]$ where θ is the *median* of $|r_t|$ on the training data. Since these methods were designed for classification, no guidance exists for thresholding, and the median is the most assumption-free choice.

Evaluation protocol. We run $N = 10{,}000$ Monte Carlo trials per scenario, each using a distinct random seed. For each trial we record whether an alarm fires, its index, and whether it occurred before or after the true drift onset, as well as distance from the true changepoint for PITMonitor. We report:

- **TPR**: fraction of trials with a true-positive alarm (alarm fired after drift onset).
- **FPR**: fraction of trials with a false alarm (alarm fired before drift onset).
- **Mean detection delay**: mean number of samples between the true drift onset and the alarm, over true-positive trials only.

5.2. Results

Table 1 presents the full results. Figure 1 visualizes TPR and FPR per method and scenario, Figure 2 shows a representative single-run monitoring trace for the GSG scenario, and Figure 3 shows the distribution of detection delays over true-positive trials. The null window is drawn from the same pre-drift distribution for all three scenarios, so each detector's FPR is near-identical across scenarios by construction.

Type I error control. PITMonitor achieves FPR of 3.8% across all scenarios, consistent with the nominal $\alpha = 0.05$ guarantee in Theorem 1. In this benchmark, ADWIN yields 0.9% empirical FPR on the 2,500-sample null window, but this number is specific to the chosen horizon and dataset. Among the other baselines, DDM and HDDM_A achieve 9.2% and 6.0% FPR, while KSWIN, PageHinkley, EDDM, and HDDM_W exceed 89% FPR with default settings; at these levels, high TPR is a trivial consequence of near-constant alarming rather than detection power, so their TPR and delay figures are not meaningfully comparable. This empirical FPR comparison should be read with care: FriedmanDrift is stationary before the changepoint by construction, so it does not stress cases with persistent low-level null violations (e.g., weak temporal dependence).

Table 1
Drift detection results on FriedmanDrift (10,000 trials, $\alpha = 0.05$).

	GRA			GSG			LEA		
Method	TPR	FPR	Delay	TPR	FPR	Delay	TPR	FPR	Delay
PITMonitor	96.2%	3.8%	77	96.2%	3.8%	189	96.2%	3.8%	1919
ADWIN	99.1%	0.9%	27	99.1%	0.9%	27	99.1%	0.9%	115
KSWIN	2.9%	97.1%	16	2.8%	97.2%	172	2.8%	97.2%	656
PageHinkley	0.4%	99.6%	1	0.4%	99.6%	6	0.4%	99.6%	70
DDM	90.8%	9.2%	405	90.0%	9.2%	666	22.0%	9.2%	2050
EDDM	8.7%	91.3%	344	8.7%	91.3%	562	1.6%	91.3%	1759
HDDM_A	94.0%	6.0%	60	94.0%	6.0%	168	52.5%	6.0%	2032
HDDM_W	9.4%	90.5%	15	9.4%	90.5%	46	9.4%	90.5%	586

Global recurring abrupt drift (GRA). On GRA, PITMonitor achieves 96.2% TPR with 3.8% FPR and mean delay 77 samples. ADWIN attains higher TPR (99.1%) and faster delay (27), but its false-alarm behavior here remains an empirical estimate tied to this finite monitoring window. DDM and HDDM_A reach 90.8% and 94.0% TPR respectively.

Global slow gradual drift (GSG). On GSG, PITMonitor achieves 96.2% TPR with 3.8% FPR and mean delay 189 samples. ADWIN again records higher TPR (99.1%) and faster delay (27), while DDM and HDDM_A reach 90.0% and 94.0% TPR respectively.

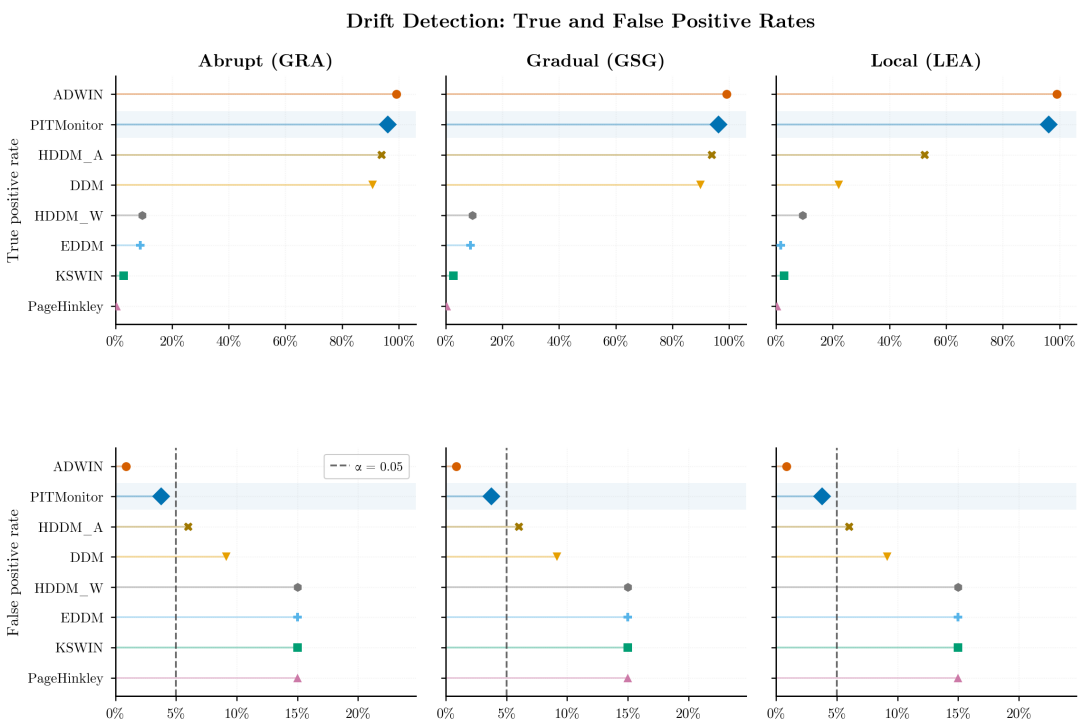

Fig 1. *TPR and FPR across all detectors and drift scenarios. The dashed line marks $\alpha = 0.05$.*

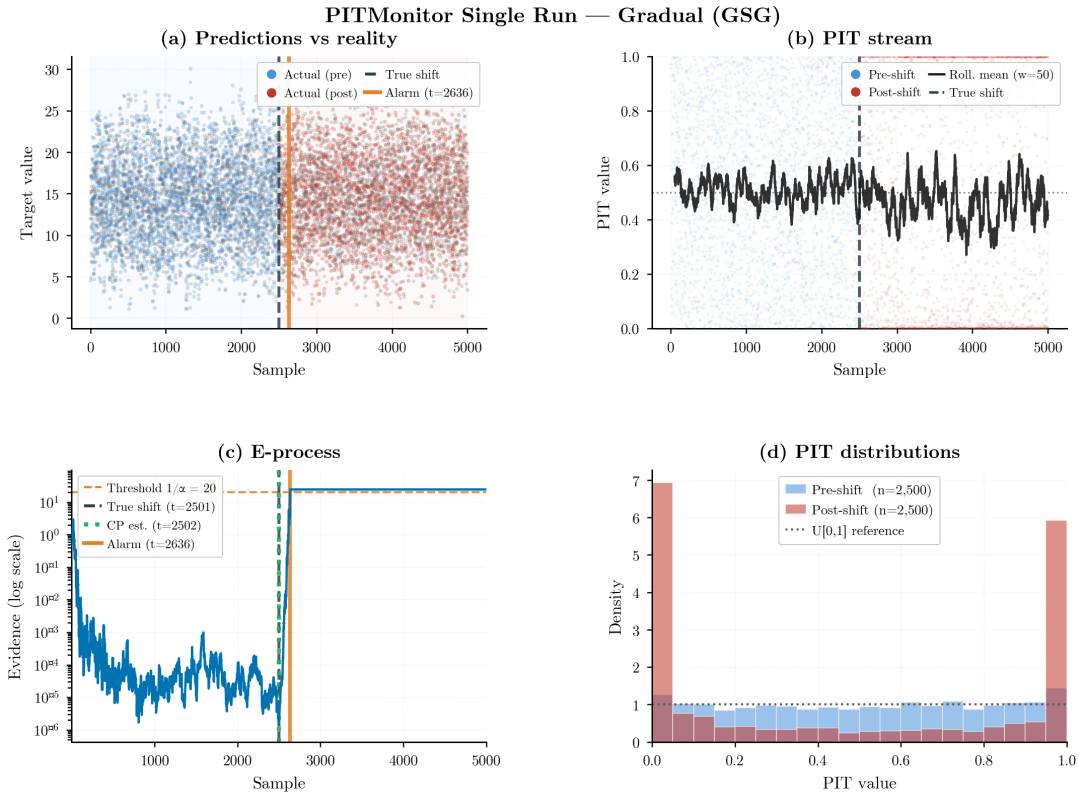

Fig 2. *Single-run PITMonitor trace. (a) Predicted vs. actual values. (b) PIT stream with rolling mean. (c) Mixture e-process on a log scale; dashed line at the threshold* $1/\alpha$*. (d) Pre- and post-shift PIT histograms.*

Local expanding drift (LEA). On LEA, PITMonitor maintains 96.2% TPR with 3.8% FPR, but mean delay rises to 1919 samples. This pattern is consistent with the expanding-drift structure: early phases induce weaker PIT distortion, so evidence accumulates slowly until later expansion stages. ADWIN detects earlier in this scenario (115 mean delay, 99.1% TPR), indicating a speed-guarantee tradeoff in this benchmark.

Detection delays. Figure 3 reports delay distributions over true-positive trials only. Accordingly, it should be interpreted jointly with TPR; a detector with low TPR may have a tight delay distribution simply because it detects only the easiest cases. PITMonitor's delay distribution is tightest under GRA, where the shift is strongest and most immediate, and widest under LEA, where the shift is weakest at onset and evidence accumulates slowly. The other detectors show similar patterns, with ADWIN achieving the fastest detection among detectors with a TPR > 90%.

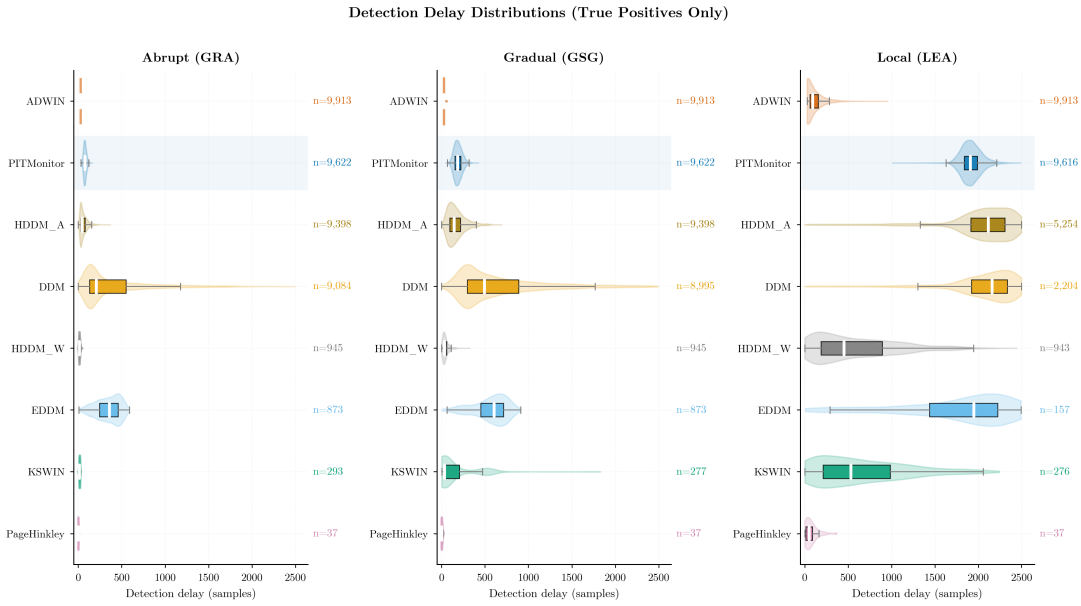

Fig 3. *Detection delay distributions (true-positive trials only) across all detectors and drift scenarios. Violins show the full delay distribution, while the white bar marks the median.*

Changepoint localization. On GRA and GSG, mean absolute error is 1.2 and 6.8 samples, indicating accurate localization under global drift. On LEA, mean error increases to 1839.3. This is expected because LEA induces multiple transition points during expansion, violating our single-changepoint localization assumption; the estimator therefore tends to select the split with strongest post-split non-uniformity, which typically occurs in later expansion phases (near offset 1,666) rather than at initial onset. In this regime, MAE should therefore be interpreted as a mismatch to a single-reference onset, not necessarily as failure to identify the most operationally relevant transition phase.

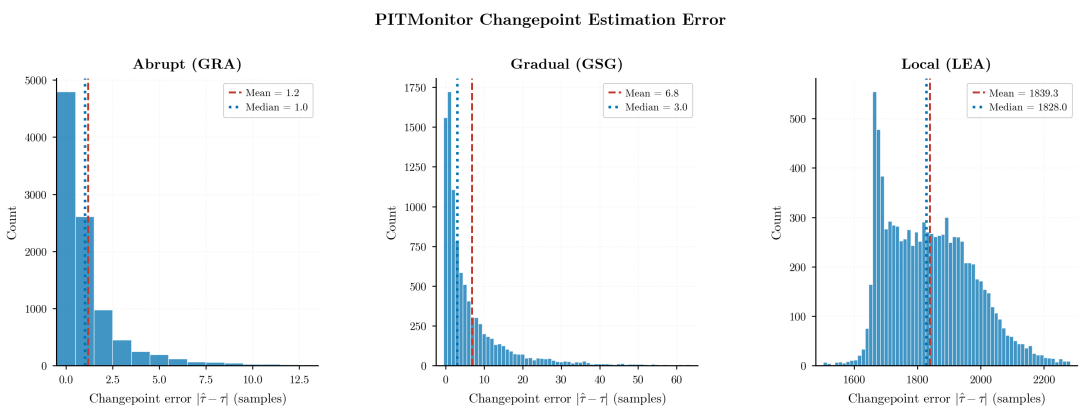

Fig 4. *Distribution of PITMonitor changepoint estimation error $|\hat{\tau} - \tau|$ across true-positive trials.*

6. Discussion

Scope. PITMonitor is intended for continuous monitoring of deployed probabilistic models when false alarms are costly, monitoring duration is open-ended, and calibration drift (not only accuracy drift) is operationally important. For one-time calibration checks or accuracy-only monitoring without need of false-alarm control or changepoint localization, standard diagnostics remain appropriate.

Limitations. *Expanding drift.* PITMonitor detects LEA drift with 96.2% TPR but with 1919 mean delay, reflecting slower evidence accumulation when only a subset of features has shifted.

Calibration improvements. PITMonitor detects any shift in the PIT distribution and does not distinguish between deterioration and improvement in calibration. A model that becomes better calibrated mid-stream may also trigger an alarm. In practice, the direction of drift can be assessed by inspecting the PIT histogram following an alarm. This is also true of two-sided baselines such as ADWIN and KSWIN, but unlike all `river` baselines — which expose only a binary alarm flag — PITMonitor's histogram-based changepoint localization provides a direct and interpretable link between the alarm and the underlying distributional change.

Changepoint localization on expanding drift. Localization is less reliable under sequential expansion: the estimator targets the phase of strongest distributional change, which can lag the true onset. This interpretation is supported by additional multi-step localization checks reported in Appendix A.

Detection delay vs. FPR control. Anytime-valid control can increase delay relative to methods without stream-level guarantees. In practice, α should be set against long-run false-alarm tolerance rather than treated as a fixed convention.

Exchangeability assumption. PITMonitor assumes PITs are i.i.d. from a fixed distribution under the null and non-uniform from a fixed distribution under the alternative.

Mild but persistent null violations due to non-stationarity can in principle inflate long-run alarms. In our FriedmanDrift experiments this did not produce extreme empirical FPR, but the benchmark is stationary pre-drift by design, so this robustness requires broader evaluation on dependent and non-stationary streams.

For univariate sequential probabilistic forecasts, correct conditional calibration — that is, correctly specifying $F_t(\cdot \mid \mathcal{F}_{t-1})$ — implies PITs are i.i.d. by construction. In this regime, PIT autocorrelation is diagnostic of misspecification (e.g., omitted lag structure or ignored volatility dynamics), so alarms are informative rather than spurious.

A separate caveat concerns PIT construction rather than detector validity: if PITs are computed from a marginal predictive that does not represent the full conditional $F_t(\cdot \mid \mathcal{F}_{t-1})$, they may be marginally uniform yet serially dependent. In that mismatched setting, exchangeability alarms can be inflated without a genuine change in the marginal PIT distribution.

The power statements in Section 4.1 should therefore be read by assumption class: Proposition 3 quantifies finite-sample mean gain under the same i.i.d. post-shift model. These assumptions are appropriate for correctly specified conditional forecasts; they do not provide finite-sample power guarantees for dependent PIT streams generated outside that regime.

Practitioners should check autocorrelation in PITs (e.g., Ljung-Box or ACF diagnostics); a practical mitigation is thinning the PIT sequence, trading reduced dependence for slower detection.

Persistence requirement. PITMonitor reacts to accumulated evidence, not single-step anomalies. Under non-exchangeability, the adaptive histogram attempts to track the current concentration pattern of p-values; when tracking is accurate, expected log-growth is positive (i.e., $\mathbb{E}[\log e_t] > 0$) and cumulative evidence increases until crossing the alarm threshold. If deviations are brief, rapidly reversing, or drift faster than the estimator can track, growth is reduced or intermittent, so alarms may be delayed or absent.

Practical recommendations. *Histogram resolution.* The number of bins B controls the bias-variance tradeoff in the density estimator. Smaller B is more stable but slower to adapt; larger B adapts faster at the cost of more variance in the estimated density and should be scaled with the expected number of monitoring samples. Additionally, during the early monitoring phase (roughly the first B observations), the histogram is dominated by the pseudocounts and e-values are nearly 1 regardless of the data, yielding low sensitivity. Practitioners expecting drift immediately after deployment may wish to use a smaller B or include a warm-start period before monitoring begins.

Structural comparison to `river` baselines. *Parameter simplicity.* In contrast to PITMonitor's two parameters (α bounding stream-level FPR, B setting histogram resolution), the `river` baselines expose many internal sensitivity parameters whose relationship to stream-level FPR is indirect, horizon-dependent, and formally unguaranteed. Aligning these methods to a target stream-level FPR typically requires advance knowledge of the monitoring window or empirical calibration.

Asymptotic behaviour of the e-process

Propositions **??** and 5 establish that under i.i.d. non-exchangeable post-change PITs, both the component and mixture e-processes grow without bound almost surely, so an alarm occurs with probability one. These results describe the asymptotic behaviour of the monitoring e-process under idealised conditions, but do not characterize how quickly evidence accumulates.

Calibration specificity. PITMonitor operates on the full PIT distribution and can therefore detect shifts affecting any distributional moment (location, scale, skewness, etc.), as long as these induce a distributional shift in the PIT stream. The `river` baselines are primarily tuned to mean-like shifts in monitored signals and are therefore not sensitive to broader notions of calibration drift.

Changepoint localization. PITMonitor provides practical changepoint localization, enabling actionable decisions about historical inference. None of the `river` baselines provide changepoint estimates.

Benchmark context. FriedmanDrift is `river`'s own synthetic benchmark; the strong performance of library-default parameters on this benchmark should be interpreted in that context.

7. Conclusion

We presented PITMonitor, an anytime-valid method for monitoring calibration of deployed probabilistic models. By monitoring for distributional shifts in probability integral transforms

via a mixture e-process, PITMonitor enables continuous monitoring without inflating Type I error, regardless of when or why monitoring stops.

Experiments on three FriedmanDrift scenarios, with baselines at library defaults, show that PITMonitor attains competitive TPR on global drift (96.2% vs. ADWIN's 99.1%) while keeping empirical FPR under the target level (3.8% at $\alpha = 0.05$). On local expanding drift, PITMonitor retains comparable TPR but with substantially larger delay, reflecting the slower evidence accumulation when only a subset of features has shifted.

Three structural advantages persist beyond the numerical comparison. First, PITMonitor's FPR guarantee is horizon-independent: PITMonitor provably satisfies $\mathbb{P}(\text{ever alarm} \mid H_0) \leq \alpha$ regardless of deployment duration, and, under idealised post-change conditions, an alarm occurs almost surely under H_1 for both the component and mixture e-processes (Propositions **??** and 5). Second, PITMonitor targets calibration specifically: by operating on probability integral transforms rather than residuals or error rates, it is sensitive to changes in predictive uncertainty that leave accuracy metrics unchanged and would be missed by error-rate or residual based detectors. Third, it provides actionable changepoint estimates that identify how far back model outputs were corrupted, directly informing decisions about how much historical inference to distrust or recompute.

In practice, PITMonitor is most suitable when formal long-run error control, changepoint localization, and calibration-specific monitoring are priorities. When gradually expanding drift is anticipated, combining PITMonitor with a faster residual-based detector can improve early warning at the cost of weaker long-run false-alarm guarantees.

Future work has four directions. First, for multivariate outputs, a practical baseline is to reduce each prediction-outcome pair to a scalar PIT and reuse the same monitoring pipeline. A fuller extension is to use a Rosenblatt chain, i.e., transform each component through sequential conditional CDFs $F(y_t^{(j)} \mid y_t^{(1:j-1)}, \mathcal{F}_{t-1})$ so that, under correct joint specification, the transformed sequence is i.i.d. uniform. The challenge is that this requires access to the full conditional factorization (or an accurate approximation), which is often unavailable for implicit or high-dimensional models. Second, we aim to improve detection power and changepoint localization under partial distributional shifts. Third, we aim to couple detection with automatic post-alarm recalibration: after an alarm, estimate the post-alarm empirical PIT density $\hat{q}(z)$ and reweight the predictive density as $\hat{f}(y) = p(y)\,\hat{q}(z)^{\lambda(e)}$, where $\lambda(e)$ is a shrinkage weight that decreases with uncertainty and is learned from e-process evidence. This would tie correction strength directly to detected miscalibration severity, yielding conservative updates under weak evidence and stronger recalibration when evidence accumulates. Fourth, a precise characterisation of p-value non-uniformity under H_1 — establishing rigorously that $G \neq F$ guarantees $p_{\tau+k} \nsim \text{Uniform}(0,1)$ for all $k \geq 1$ — would complete the theoretical picture; the main obstacle is ruling out cases where $G \circ F^{-1}$ matches the moments of $\text{Uniform}(0,1)$ up to the degree determined by the reference size.

Code Availability. `https://github.com/tristan-farran/pitmon`.

Appendix A: Additional Validation Checks

To keep the main text focused on the core benchmark and theorem-driven claims, we report supplementary checks here. These diagnostics are not used for model or detector tuning and are not part of the primary comparative evaluation; they are included to document robustness and interpretation boundaries.

Finite-time proposition sanity check. The finite-sample mean formula from Proposition 3 is numerically consistent with Monte Carlo estimates for representative aligned and misaligned priors, with maximum absolute discrepancies of 0.007 (aligned prior) and 0.009 (misaligned prior). This check is a numerical consistency test of the proved expression rather than independent additional evidence.

Multi-step localization interpretation check. In a staged step-drift stream, changepoint localization is substantially closer to the maximal-intensity phase than to initial onset: MAE to onset is 1808.2 versus 265.6 to the strongest phase, and 9,629 of 10,000 runs are closer to the strongest phase reference. This result is used to interpret localization behavior under expanding/staged drift, not as a primary benchmark metric.

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